



TECHNICAL PROPOSAL FOR PHARMACY E-COMMERCE WEBSITE DESIGN & DEVELOPMENT

Prepared For	Prepared By	Date
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INTRODUCTION

Niche Web Creation presents this comprehensive proposal for the development of a world-class pharmacy e-commerce platform that will position your pharmacy as the premier digital health destination in Kenya. This platform will borrow leaf from goodlife.co.ke but offer superior functionality combining a fast, trustworthy online pharmacy with genuine healthcare value-add: verified prescription handling, confidential consultation booking, community health education, and enterprise-grade data protection.

Our proposed solution integrates advanced e-commerce functionality with specialized healthcare features including official prescription submission, confidential medical consultation booking, bulk SMS notifications, and robust data security compliance frameworks. The platform will also serve as a community health education resource, delivering trusted wellness content to your customers.

Built on modern, scalable architecture with a mobile-first approach, the platform will deliver exceptional performance across all devices while supporting local payment gateways including M-Pesa, and or PesaPal, DPO, and IntaSend for seamless one-step checkout.

The project will be delivered within a 3-month timeframe, followed by 5 months of complimentary support, comprehensive end-user training, and a structured handover process to ensure your team is fully empowered to manage the platform.

PROJECT OVERVIEW & OBJECTIVES

Our primary objective is to develop an advanced pharmacy e-commerce platform that combines commercial excellence with healthcare compliance and community education. The platform will be engineered to exceed the standards set by current market leaders while delivering a superior user experience.

Key Objectives

1. Build a feature-rich e-commerce platform superior to goodlife.co.ke in functionality, design, and user experience
2. Implement secure prescription submission and verification workflows compliant with pharmaceutical regulations
3. Enable confidential medical consultation booking with qualified healthcare professionals

4. Integrate bulk SMS for order updates, prescription reminders, and health alerts
5. Ensure full compliance with data protection regulations (Data Protection Act 2019) and pharmaceutical laws
6. Create a health education hub with wellness articles, prevention guidelines, and healthy living tips
7. Deliver intuitive one-step checkout with multiple local payment gateway options
8. Implement strict Role-Based Access Control (RBAC) for all backend operations

Our 10+ years of experience and approach aligns perfectly with your vision. We specialize in the development of modern and custom ecommerce platforms that are not only robust and scalable but also delivered with unparalleled efficiency and precision.

SCOPE OF WORK

The following sections detail the comprehensive scope of features and functionalities to be developed for the pharmacy e-commerce platform.

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E-Commerce Platform Features

Advanced Product Filtering and Search

The platform will feature an intelligent search and filtering system allowing customers to find products by category, brand, condition, symptoms, price range, availability, and ratings. Auto-suggestions, spell correction capabilities will enhance discoverability.

Product Showcase

Each product will have a detailed showcase page featuring high-resolution images, comprehensive descriptions, active ingredients, dosage instructions, side effects, customer reviews, related products, and stock availability indicators.

Intuitive One-Step Checkout

A streamlined single-page checkout process will minimize cart abandonment. Customers can review their order, select delivery options, choose payment methods, and complete purchase with minimal friction. Guest checkout will be available for first-time buyers.

Secure Customer Login & Guest Checkout

The platform will support both registered user accounts with full order history and guest checkout for quick purchases. Social login via Google will be available. Account management will include profile editing, saved addresses, prescription history, and consultation records.

Prescription Management System

A dedicated prescription module will enable customers to upload official prescriptions through multiple channels:

1. Digital upload via mobile camera or file attachment (PDF, JPG, PNG)
2. Manual entry with prescriber details and medication information
3. Direct prescription transfer from participating healthcare providers

All prescriptions will be verified by authorized pharmacists through a secure backend workflow before order processing. The system will maintain prescription history, track refill dates, and send automated reminders when refills are due.

Prescription Status Tracking

Customers can track their prescription status in real-time: Submitted, Under Review, Approved, Processing, Ready for Pickup/Delivery, and Completed. SMS and email notifications will be sent at each status change.

Medical Consultation Module

The confidential medical consultation system will enable customers to request and book consultations with qualified healthcare professionals:

1. Online consultation request form with symptom description and medical history
2. Appointment scheduling with available healthcare professionals
3. Video, audio, and chat-based consultation options
4. Secure messaging for follow-up questions
5. Digital prescription issuance post-consultation

All consultations will be conducted through encrypted channels with full confidentiality protection. Consultation records will be accessible only to the patient and the attending healthcare professional.

Community Health & Education Hub

The platform will feature a comprehensive health education section designed to build brand authority and community trust:

1. Healthy Living Tips - Regular articles on nutrition, exercise, mental wellness, and lifestyle
2. Disease Prevention Guidelines - Evidence-based prevention measures for common conditions
3. Medication Guides - Proper usage, storage, and disposal information for medications
4. Seasonal Health Alerts - Timely warnings about disease outbreaks and preventive measures
5. Expert Health Blog - Articles authored by healthcare professionals
6. FAQ Section - Common health and medication questions answered
7. Health Newsletter - Subscription-based weekly health updates via email and SMS

Content will be SEO-optimized to drive organic traffic and establish the pharmacy as a trusted health information resource.

Communication & Notifications

Bulk SMS Integration

The platform will integrate with bulk SMS providers to enable:

1. Order confirmation and status updates
2. Prescription ready notifications
3. Medication refill reminders
4. Health tips and seasonal alerts
5. Promotional campaigns and loyalty rewards

Email Notifications

Automated email notifications for order confirmations, shipping updates, consultation reminders, health newsletters, and promotional offers.

Technical Architecture

The platform will be built using a modern, scalable technology stack designed for performance, security, and maintainability:

Component	Technology
Backend Framework	Yii2 Advanced (PHP)
Frontend Framework	Bootstrap 5 + jQuery
Database	MySQL 8.0 with full-text search
Caching	Redis + Query caching
Search Engine	Elasticsearch for product search
Queue/Background Jobs	Redis + Yii2 Queue
File Storage	AWS S3 / Local with CDN
API Layer	RESTful API + OAuth2 authentication

The architecture follows a modular, service-oriented design that enables independent scaling of components and easy integration of future features. All code will be version-controlled with comprehensive documentation.

DATA SECURITY, PRIVACY & COMPLIANCE

Data security and patient privacy are paramount in healthcare technology. The platform will implement comprehensive security measures to protect sensitive information:

Security Features

- SSL/TLS encryption for all data in transit (HTTPS everywhere)
- AES-256 encryption for sensitive data at rest
- Secure password hashing with bcrypt
- Two-factor authentication (2FA) for admin and pharmacy staff accounts
- SQL injection and XSS attack prevention
- CSRF token protection on all forms
- Rate limiting to prevent brute force attacks
- Automated security scanning and vulnerability assessments
- Regular automated backups with off-site storage
- Intrusion detection and monitoring

Compliance

- Full compliance with Kenya Data Protection Act 2019
- Pharmacy and Poisons Board regulations adherence
- GDPR-ready architecture for future international expansion
- Clear data privacy policy and terms of service
- User consent management for data collection and marketing communications
- Right to access, rectification, and erasure of personal data

Role-Based Access Control (RBAC)

Strict RBAC will govern all backend operations with the following role hierarchy:

- Super Administrator - Full system access and configuration
- Pharmacy Manager - Product management, order oversight, staff management, reporting
- Licensed Pharmacist - Prescription verification, consultation management
- Sales/Customer Service - Order processing, customer support, enquiry management
- Content Manager - Health articles, promotions, banner management

PAYMENT GATEWAY INTEGRATION

The platform will integrate with leading local and international payment providers to ensure maximum convenience and conversion:

Gateway	Features
M-Pesa	STK Push, Pay Bill, Buy Goods, B2C refunds
PesaPal	Card payments, mobile money aggregation, recurring billing
DPO Pay	Multi-currency, card payments, mobile wallets across Africa
IntaSend	Instant payouts, collection APIs, wallet transfers
Cash on Delivery	Optional COD with delivery verification
Card Payments	Visa, Mastercard via PesaPal/DPO with 3D Secure

All payment processing will be PCI-DSS compliant with tokenized card storage. M-Pesa STK Push will be the primary mobile payment method, complemented by Pay Bill and Buy Goods options.

ANALYTICS & REPORTING

Comprehensive analytics and reporting will provide actionable business insights:

- Sales Dashboard - Revenue, orders, average order value, conversion rates
- Product Analytics - Best sellers, inventory turnover, profit margins
- Prescription Analytics

Integration with Google Analytics 4 and Google Tag Manager will enable detailed tracking of user behavior and marketing attribution.

MOBILE RESPONSIVENESS

With over 85% of Kenyan internet users accessing the web via mobile devices, the platform will be built with a mobile-first design philosophy:

- Responsive design optimized for all screen sizes (320px to 1920px+)
- Touch-friendly interface with appropriate tap targets
- Offline browsing of previously loaded pages
- Add to Home Screen functionality
- Push notification support for order updates and promotions
- Mobile-optimized checkout with M-Pesa STK Push integration
- Simplified mobile navigation with bottom tab bar

PROJECT TIMELINE

The project will be delivered over a 3-month period structured in agile sprints:

Phase	Duration	Deliverables
Month 1	Discovery & Design	Requirements finalization, UI/UX design, database architecture, project setup
Month 1-2	Core Development	E-commerce engine, user management, product catalog, payment integration

Phase	Duration	Deliverables
Month 2	Healthcare Features	Prescription system, consultation module, health education hub
Month 2-3	Integration & Testing	SMS integration, security audit, performance optimization, UAT
Month 3	Deployment & Training	Production deployment, staff training, data migration, launch

Each sprint will include planning, development, testing, and client review phases. Weekly progress meetings will keep stakeholders informed and enable rapid feedback incorporation.

INVESTMENT

We believe in transparent, value-based pricing. For a fully customized, advanced website as outlined above, we provide a fixed project cost. This ensures no surprises.

DESCRIPTION		AMOUNT(KSh)
Ecommerce Platform Development		KSh150,000
	SUBTOTAL	KSh150,000
	TOTAL	KSh150,000

The investment covers all development costs, testing, deployment, training, and 5 months of complimentary post-launch support. Domain registration and premium hosting are included for the first year.

Payment Schedule

- **Deposit:** KSh50, 000 upon contract signing
- **Mid-project:** KSh50, 000 (after development completion).
- **Final:** KSh50, 000 - upon successful launch and training completion.

MAINTENANCE & SUPPORT SLA

Following the 5-month complimentary support period, ongoing maintenance and support will be available at 10% of the total project cost per month. This covers:

- 24/7 system monitoring and uptime alerts
- Priority bug fixes and issue resolution
- Security patches and updates
- Monthly platform performance reports
- Weekly automated backups with quarterly disaster recovery testing
- Content updates (up to 10 changes per month)
- Technical support via email and phone during business hours
- Quarterly feature enhancement planning sessions

TERMS & CONDITIONS

The following terms and conditions govern this proposal and subsequent engagement:

Intellectual Property

All source code, designs, documentation, and deliverables created specifically for this project will become the exclusive property of the Client upon full payment. Niche Web Creation retains the right to showcase the project in its portfolio.

Confidentiality

Both parties agree to maintain strict confidentiality regarding proprietary information, business data, customer information, and trade secrets disclosed during the engagement. This obligation survives contract termination.

Warranties

Niche Web Creation warrants that all deliverables will be free from material defects for a period of 5 months from launch. This warranty covers bug fixes at no additional cost. The warranty does not cover issues arising from third-party service changes, unauthorized modifications, or force majeure events.

Limitation of Liability

Niche Web Creation's total liability shall not exceed the total contract value. Neither party shall be liable for indirect, consequential, or incidental damages. The platform is provided as-is with respect to third-party integrations subject to their respective terms of service.

Termination

Either party may terminate the engagement with 30 days written notice. In the event of termination, we will deliver all completed work and source code proportional to payments received.

Data Handling

All customer data, prescription information, and consultation records are the exclusive property of the Client. Niche Web Creation will not access, use, or disclose any data except as required for development, maintenance, and support purposes. Upon termination, all data will be returned or securely destroyed as directed by the Client.

WHY CHOOSE NICHE WEB CREATION?

- **Proven Expertise:** Extensive experience with product-focused websites and an understanding of the catalogue based website industry.
- **Collaborative Approach:** We believe in responsive and communicative partnerships, working closely with your team.
- **Comprehensive Support:** We provide training and robust post-launch support.
- **Full Ownership:** All source code and assets will be handed over to the client upon completion.
- **Scalable Solutions:** Our technical architecture is designed to grow with your business, easily accommodating new products, markets, and features as needed.